Final Design Review (FDR)

Purpose of the FDR

The Final Design Review (CDR) is a summative presentation held towards the end of the project. Its purpose is to:

- Show your final concept and why it meets mission requirements
- Demonstrate the sensitivity of the final concept to key design drivers
- State modelling assumption limitations and risks for future progress
- Demonstrate effective teamworking principles
- Provide a final opportunity for feedback prior to the submission of the FEDR.

The FDR is **not** about presenting a detailed account of each discipline. It is about demonstrating the ‘big picture’, showing clear thinking, well-justified decisions, and an awareness of the key design drivers identified during the project.

Presentation Format

Each group will have up to 12 minutes to present. This will be followed by a Q&A session with the assessment panel. As a guide, 7-12 clear slides are usually appropriate for a 12-minute presentation.

What Your Presentation Should Cover

The primary aim of the FDR is to present your final optimal design concept, supported by system-level reasoning and key sensitivities. You have limited time in your presentation, so you will not be able to talk about each discipline in detail. Therefore, your presentation should focus on the big picture, key decisions, and the most critical engineering challenges for your chosen concept. As an example, your presentation may aim to cover the following topics:

- Design Task and Mission Overview
    - Briefly describe the design task, your chosen ‘design mission’, and highlight mission-critical requirements for your aircraft
- Final-Concept
    - Describe your chosen concept and highlight the key potential benefits of this concept in relation to the customer or mission requirements. (comment on other concepts considered)
- Optimisation and Sensitivity studies
    - Summarise the critical design parameters and their effect at the aircraft level.
    - Compare DOC-optimal vs climate-optimal solutions if relevant.
- Key Technical Challenges
    - Identify the main engineering challenges encountered when developing the sizing tool. These may be intra- or inter-disciplinary challenges
    - Focus on challenges that significantly affect the overall aircraft design
- Project Management:
    - Discuss team management strategy and comment on how project management was improved between each sprint.
    - Identify key risks for the “hypothetical” further development of your concept by a future engineering team.

What the Panel is Looking For

Given the limited time, we are not expecting equal 'airtime' for each discipline. Instead, a strong FDR will show you:

- Understand the design problem and mission
- Have made reasonable assumptions and justifiable decisions
- Have identified the main technical and integration challenges
- Show that the team has focused effort where it matters most for a successful outcome of the project

Preparing for the Q&A

After each presentation, there will be a Q&A session with the assessment panel. Each team member (not just those presenting) must be prepared to answer questions. Example questions may be on topics such as assumptions, uncertainties, key sensitivities, feasibility, and integration challenges. It is ok to include an "appendix" to your presentation (additional slides) that you may wish to refer to during questions.

Presenters and Attendance

There are two opportunities for group presentations during this project: the CDR and FDR (Final Design Review).

- A maximum of four team members may present at the CDR or the FDR
- Every team member must present at either the CDR or the FDR
- All team members must be in attendance for both the CDR and FDR, and should be prepared to answer questions during the Q&A session

There is no requirement for presenters to speak for equal amounts of time — divide speaking roles in a way that best supports a clear and coherent presentation.

Final Advice

The goal of the FDR is to convince the panel that you have a clear understanding of the design challenge and have developed a suitable design, built upon clear assumptions and sound engineering judgment. To achieve this, it is important to:

- Be clear rather than comprehensive – don't rush through too much content
- State your assumptions – a successful team will know the limitations of their modelling assumptions
- Use simple diagrams and sketches where possible – help the panel visualise what you are talking about

Overall, if you show good engineering judgement and an honest reflection of the challenges, you are on course to do well in this unit.

Times, dates, and locations of FDR presentations, as well as deadlines, will be provided on the main BB page.